

Chapter 02: Complex numbers

1) Algebraic form: (Cartesian form)

Definition: The field of complex numbers, denoted as:

$$\mathbb{C} = \{z = x + iy \mid x, y \in \mathbb{R}, i^2 = -1\}$$

The algebraic form a complex number z is: $z = x + yi$

x is the real part $\operatorname{Re}(z) = x$

y is the imaginary part $\operatorname{Im}(z) = y$

if $x=0$: z is real

if $y=0$: z is pure imaginary

Operations:

$$z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$$

$$z_1 - z_2 = (x_1 - x_2) + i(y_1 - y_2)$$

$$\frac{1}{z} = \frac{1}{x + iy} = \frac{x - iy}{x^2 + y^2} = \frac{x}{x^2 + y^2} - i \frac{y}{x^2 + y^2}, \quad z \neq 0$$

$$\alpha z = \alpha(x + iy) = \alpha x + i\alpha y \quad (\alpha \in \mathbb{R})$$

Conjugate:

conjugate of z denoted as $\bar{z}$ is: $\bar{z} = x - yi$

and its properties are:

$$\overline{\bar{z}} = z, \quad \overline{(z + z')} = \bar{z} + \bar{z'}, \quad \overline{z \cdot z'} = \bar{z} \cdot \bar{z'}, \quad \overline{\left(\frac{z}{z'}\right)} = \frac{\bar{z}}{\bar{z'}} \text{ where } z' \neq 0$$

$$z \cdot \bar{z} = x^2 + y^2$$

Modulus:

The modulus of z denoted as $|z|$ is $|z| = \sqrt{x^2 + y^2}$

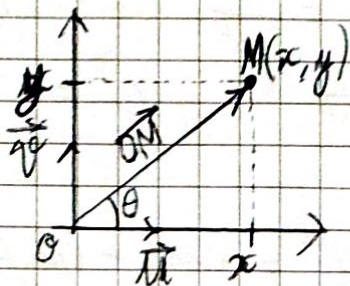
and its properties are: $|z| = 0 \Leftrightarrow z = 0$, $\left|\frac{1}{z}\right| = \frac{1}{|z|}, z \neq 0$

$$|z \cdot z'| = |z| \cdot |z'|, \quad |z| = |\bar{z}| = |-z| = \left|-\frac{z}{z}\right| = \frac{|z|}{|z|}, \quad |z^n| = |z|^n$$

$$|z + z'| \leq |z| + |z'|, \quad |z - z'| \geq ||z| - |z'||$$

Graphical representation of a complex number:

To every $z \in \mathbb{C}$ in the plane equipped with orthonormal coordinate $(O, \vec{u}, \vec{v})$ we associate its image $M(x, y)$ and any vector image $\vec{OM}$ where z is called the affix of M and affix of $\vec{OM}$.



Argument:

denoted as $\arg(z)$ is the angle $\theta = (\vec{u}, \vec{OM})$, or $\arg(z) = \theta + 2k\pi$ ($k \in \mathbb{Z}$)

$$\cos \theta = \frac{x}{|z|}, \sin \theta = \frac{y}{|z|}, \tan \theta = \frac{y}{x}$$

its properties:

$$\arg(z) = -k\pi, k \in \mathbb{Z} \Leftrightarrow z \in \mathbb{R}$$

$$\arg(z) = \frac{\pi}{2} + k\pi, k \in \mathbb{Z} \Leftrightarrow z \text{ is pure imaginary}$$

$$\arg(\bar{z}) = -\arg(z)$$

$$\arg\left(\frac{1}{z}\right) = -\arg(z)$$

$$\arg(-z) = \arg(z) + \pi$$

$$\arg(\alpha z) = \arg(z), \forall \alpha \in \mathbb{R}_+^*$$

$$\arg(z \cdot z') = \arg(z) + \arg(z')$$

$$\arg\left(\frac{z}{z'}\right) = \arg(z) - \arg(z')$$

$$\arg(z^n) = n \arg(z)$$

Exponential form:

of z is: $z = |z|(\cos \theta + i \sin \theta)$

properties:

$$z_1 \cdot z_2 = |z_1| |z_2| (\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2))$$

$$z^n = |z|^n (\cos(n\theta) + i \sin(n\theta))$$

Exponential form:

we have $e^{i\theta} = \cos\theta + i\sin\theta$ so that:

$$Z = |Z| e^{i\theta}$$

Properties:

$$e^{i(\theta+\theta')} = e^{i\theta} \cdot e^{i\theta'}$$

$$e^{-i\theta} = \frac{1}{e^{i\theta}}$$

$$e^{i(\theta-\theta')} = \frac{e^{i\theta}}{e^{i\theta'}}$$

$$\overline{e^{i\theta}} = e^{-i\theta} = \frac{1}{e^{i\theta}}$$

$$(e^{i\theta})^n = e^{in\theta}$$

$$e^{i\theta} = 1 \Rightarrow \theta = 2K\pi \quad (K \in \mathbb{Z})$$

Euler formula:

$$e^{i\theta} = \cos\theta + i\sin\theta$$

$$+ e^{-i\theta} = \cos\theta - i\sin\theta$$

$$\downarrow e^{i\theta} + e^{-i\theta} = 2\cos\theta$$

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

$$e^{i\theta} = \cos\theta + i\sin\theta$$

$$- e^{-i\theta} = -\cos\theta + i\sin\theta$$

$$\downarrow e^{i\theta} - e^{-i\theta} = 2i\sin\theta$$

$$\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

Möbius formula:

$$(\cos\theta + i\sin\theta)^n = (\cos(n\theta) + i\sin(n\theta)) \quad \forall n \in \mathbb{Z}, \theta \in \mathbb{R}$$

The n^{th} root of a complex number:

$$Z = z^n$$

$$|z| e^{i\theta_2} = (|Z| e^{i\theta_1})^n$$

$$|z| e^{i\theta_2} = |Z|^n e^{i(n\theta_1)}$$

$$\wedge \arg(z) = \arg(Z)$$

$$\arg(|z| e^{i\theta_2}) = \arg(|Z|^n e^{i(n\theta_1)})$$

$$|z| = |Z|^n$$

$$|z| = \sqrt[n]{|Z|} \quad \text{--- (1)}$$

$$i\theta_2 = i(n\theta_1 + 2K\pi) \Rightarrow i\theta_2 = i(\theta_1 + 2K\pi/n)$$

$$Z = \sqrt[n]{|Z|} e^{i(\theta_1 + 2K\pi/n)}$$

$$2K\pi + \theta_1 = n\theta_2$$

$$\theta_2 = \frac{2K\pi + \theta_1}{n}$$

$$\text{so: } Z = \sqrt[n]{|Z|} e^{i(\frac{2K\pi + \theta_1}{n})}$$